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Escaping families of continuous group actions

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This talk is about *algebraic exponentiability* for topological groups. Our aim is to explain what the categorical-algebraic conditions called *algebraic coherence* [5] and *local algebraic cartesian closedness* [7, 3] amount to in the category $\mathsf{TopGp} = \mathsf{Gp}(\mathsf{Top})$, which is already known to be a homological category [1, 2] with representable internal actions [4].

At the heart of the problem is the infamous complexity of the coproduct of topological groups, which is the free product of the underlying groups, equipped with a topology that is renowned for its intractability. Our proposed way out is to view the question from a purely categorical angle: instead of an explicit construction of that coproduct, we use only its universal property. An 'escaping family of continuous group actions' as in the title is what allows us to conveniently make that universal property fail.

Our main result extends the Kaluzhnin–Krasner universal embedding theorem [8, 6] beyond the classical case where the groups are discrete.

This is joint work-in-progress with Lennert De Baecke, David Forsman and Mark Sioen. The speaker is a Research Director of the Fonds de la Recherche Scientifique–FNRS.

Keywords: Algebraic coherence; algebraic exponentiability; homological category; Kaluzhnin–Krasner universal embedding theorem; local algebraic cartesian closedness; topological group; wreath product
Subject area: Categorical Algebra
MSC2020: 18E13, 20E22, 22A05, 54H11

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